

Python Development
Avni goel | TEN/PY/1206
assignment code 2 (day 3)

Question – Production Counter System

Code –

```
import random

target_units = int(input("Enter target units: "))
workers_per_shift = int(input("Enter workers per shift: "))
defect_rate = float(input("Enter defect rate (%): "))
total_produced = 0

for shift in range(1, 4):
    shift_produced = 0
    defects = 0
    print(f"\n--- Shift {shift} Started ---")
    for cycle in range(1, 21):
        if total_produced >= target_units:
            print("\nTarget reached. Stopping production.")
            break

        for worker in range(1, workers_per_shift + 1):
            if total_produced >= target_units:
                break

            chance = random.randint(1, 100)

            if chance <= defect_rate:
                defects += 1
```

```
        continue

    shift_produced += 1

    total_produced += 1

    if total_produced >= target_units:

        break

    productivity = shift_produced / workers_per_shift

    print(f"Items Produced : {shift_produced}")

    print(f"Defective Items: {defects}")

    print(f"Worker Productivity: {productivity:.2f} items/worker")

print("\n===== FINAL REPORT =====")

print(f"Total Good Units Produced: {total_produced}")

print(f"Production Target: {target_units}")
```

```

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        if total_produced >= target_units:
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        for worker in range(1, workers_per_shift + 1):
            if total_produced >= target_units:
                break
            chance = random.randint(1, 100)
            if chance <= defect_rate:
                defects += 1
                continue
            shift_produced += 1
            total_produced += 1
        if total_produced >= target_units:
            break
    productivity = shift_produced / workers_per_shift
    print(f"Items Produced : {shift_produced}")
    print(f"Defective Items: {defects}")
    print(f"Worker Productivity: {productivity:.2f} items/worker")
print("\n===== FINAL REPORT =====")
print(f"Total Good Units Produced: {total_produced}")
print(f"Production Target: {target_units}")

```

Output –

Enter target units: 250

Enter workers per shift: 4

Enter defect rate (%): 12.5

--- Shift 1 Started ---

Items Produced : 74

Defective Items: 6

Worker Productivity: 18.50 items/worker

--- Shift 2 Started ---

Items Produced : 68

Defective Items: 12

Worker Productivity: 17.00 items/worker

--- Shift 3 Started ---

Items Produced : 71

Defective Items: 9

Worker Productivity: 17.75 items/worker

===== FINAL REPORT =====

Total Good Units Produced: 213

Production Target: 250

```
Enter target units: 250
Enter workers per shift: 4
Enter defect rate (%): 12.5

--- Shift 1 Started ---
Items Produced : 74
Defective Items: 6
Worker Productivity: 18.50 items/worker

--- Shift 2 Started ---
Items Produced : 68
Defective Items: 12
Worker Productivity: 17.00 items/worker

--- Shift 3 Started ---
Items Produced : 71
Defective Items: 9
Worker Productivity: 17.75 items/worker

===== FINAL REPORT =====
Total Good Units Produced: 213
Production Target: 250
```